

A low-angle, upward-looking perspective of several tall skyscrapers. The buildings are dark and silhouetted against a vibrant, solid red sky. The perspective creates a sense of height and scale, with the buildings converging towards the top of the frame.

Foundation Workshop **COST MANAGEMENT**

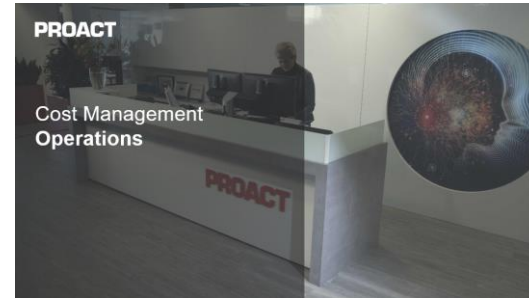
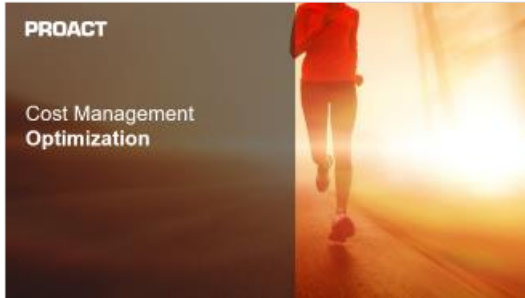
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Azure Cost Control Session

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Cloud Solution Architect

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Agenda



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Cost Management Overview



Azure versus On-Premise Cost Calculation

Azure

- Pay per use, Enterprise Agreement
- Oversized resources increase cost
- Unused resources increase cost
- Flexible sizing, upscale and downscale
- Many resources available as-a-service
- Predictable monthly/yearly cost

On-Premise

- Upfront investment, depreciate investment in years
- Oversized resources don't impact
- Unused resources don't impact
- No flexible upscale; new investment
- All resources need to be fully managed
- High investment after depreciate period

Cost Analysis in Azure

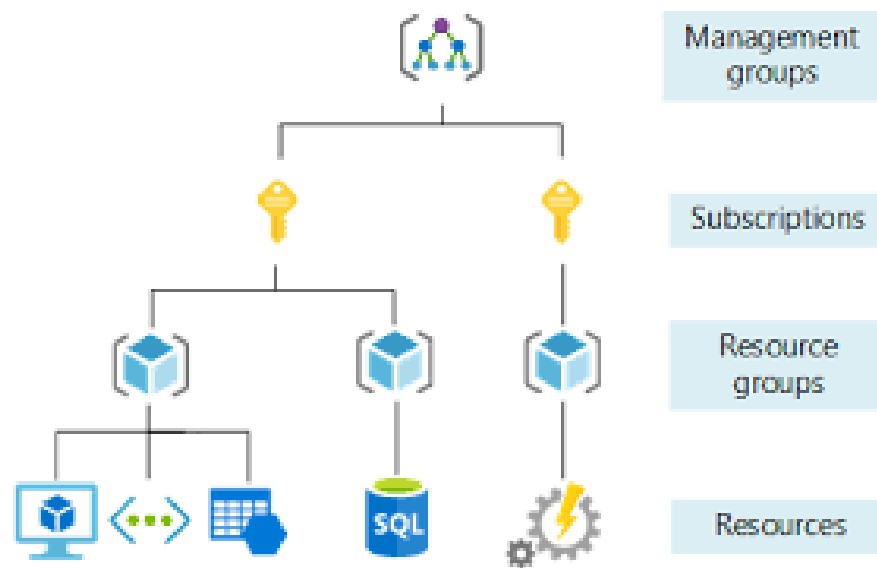
- Billing accounts support
- **Microsoft Online Services Program**
 - » Pay-as-you-Go
 - » Free Azure
 - » Visual Studio subscription
- **Enterprise Agreement**
- **Microsoft Customer Agreement**

Cost Analysis in Azure

- Billing accounts support
- Review and analyze by **Scope**
 - » Billing account scopes
- **Microsoft Online Services Program**
 - » Billing account
 - » Subscription
- **Enterprise Agreement**
 - » Billing account
 - » Department
 - » Enrollment account
- **Microsoft Customer Agreement**
 - » Billing account
 - » Billing profile
 - » Invoice section

Resource management scopes

- Management Groups
 - » Subscription
 - » Resource group
- Filter on
 - » Tags
 - » Default Meter properties
 - » Category
 - » Subcategory
 - » Partnumber
 - » ...

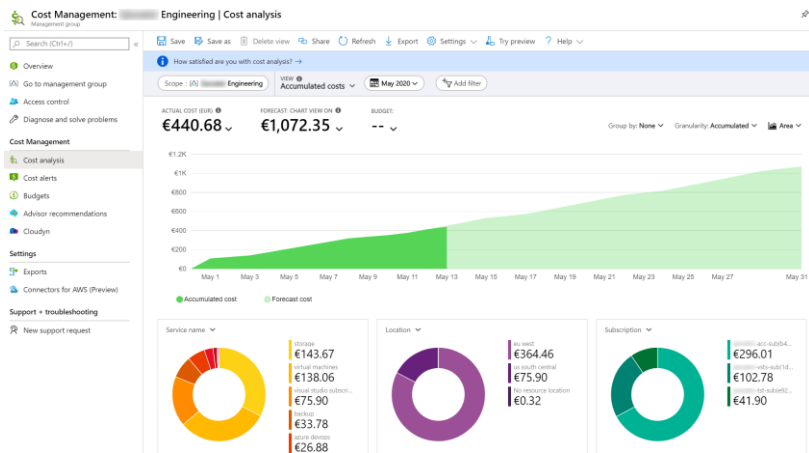
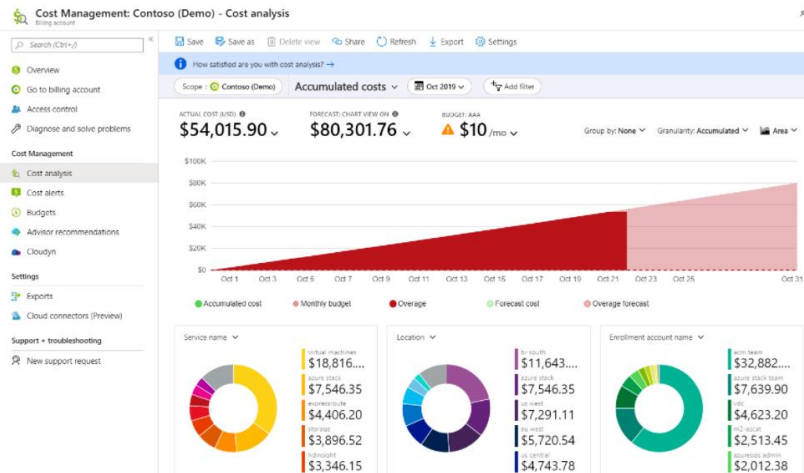


Costs & Forecast

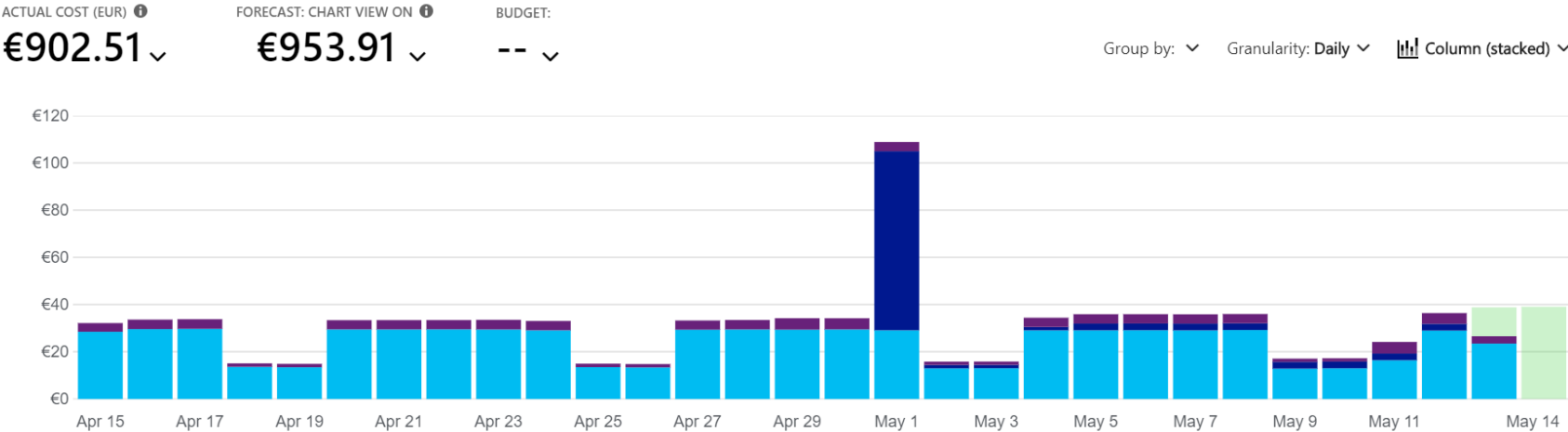
Cost Analysis

- Customize Cost view
 - » Save and share customized views with others by pinning cost analysis to Azure portal dashboard
- Multiple viewpoints
 - » Cost by Resource
- Minimal Read access in Azure required
- Enhance views by implementing Budgets and Tags
- The forecast model can predict future costs for up to a year.

Aug 31			
Accumulated cost	--		
Monthly budget	\$3,500.00		
Overage	--		
Forecast cost	\$3,500.00		
Overage forecast	\$213.11		



Cost Analysis; Daily Costs View



Additional Microsoft tools for Azure Cost analysis

- **Azure Pricing Calculator**
 - » Use this tool to estimate your up-front cloud costs.
- **Azure Migrate**
 - » Assess your current datacenter workload for insights about what's needed from an Azure replacement solution.
- **Azure Advisor**
 - » Identify unused VMs and receive recommendations about Azure reserved instance purchases.
- **Azure Retail Prices REST API overview**

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Cost Management Budgets & Alerts



Implement Budgets to get alerts when reaching threshold

- Track spend to budget
- Determine budgets (threshold level)
- Get alerts; Action Groups
 - » Email
 - » Phone/SMS
 - » Webhook trigger
- Use Billing API

How satisfied are you with budgets? →

Scope : <SubscriptionName>

Search by name All periods

NAME	SCOPE	RESET PE...	START D...	END DATE	BUDGET	CURRENT SPEND	PROGRESS
AdamTestBudg...	847e9a39-4c9...	Monthly	1/1/2019	12/31/2020	\$5.00	\$99.42	1,988.39% ...
hadarsTest	847e9a39-4c9...	Monthly	1/1/2019	12/31/2020	\$11.00	\$99.42	903.81% ...
AviTest	847e9a39-4c9...	Quarterly	2/1/2019	1/31/2021	\$1,000.00	\$99.42	9.94% ...
AviTest52	847e9a39-4c9...	Monthly	2/1/2019	1/31/2021	\$1,000.00	\$99.42	9.94% ...
Test2	847e9a39-4c9...	Monthly	2/1/2019	1/31/2021	\$1,000.00	\$99.42	9.94% ...
hadarsTest2	847e9a39-4c9...	Monthly	2/1/2019	1/31/2021	\$1,000.00	\$99.42	9.94% ...
Budget1	847e9a39-4c9...	Monthly	2/1/2019	1/31/2021	\$10,000.00	\$99.42	0.99% ...

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Cost Management Optimization

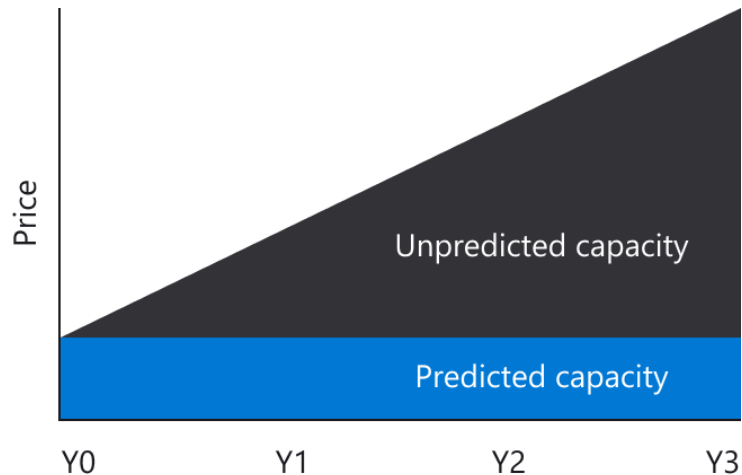


Rightsize

- **Identify right size opportunities**
 - » Use Azure Advisor for automated recommendations
 - » Use Azure Monitor for manual optimization
- **Right-size provisioned SKUs**
 - » Modify underutilized resources
- **Adjust autoscale parameters**
 - » Find usage patterns

Reserved instances

- Buy capacity upfront for predictable use
 - » 1 or 3 years
 - » Pay per month or upfront
- Scope
 - » Shared – All EA subscriptions
 - » Single subscription
 - » Single resource group
- Optimized for
 - » Instance size flexibility
 - » Capacity priority



Reserved Instances - Applicable resources

- Azure Virtual Machine Instance
- Azure Blob Storage
- Azure Cosmos DB reserved capacity
- SQL Database reserved vCore
- Azure Synapse Analytics (SQL DW)
- Azure Databricks
- App Service stamp fee
- Azure Files reserved capacity
- Azure Managed Disks
- Azure Database for MySQL
- Azure Database for PostgreSQL
- Azure Database for MariaDB
- Azure Data Explorer
- Azure Cache for Redis
- Azure Dedicated Host
- Azure VMware Solution
- Azure Data Factory
- SUSE Linux, Azure Red Hat OpenShift

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Hybrid Benefit

Use your on-premise Enterprise Licenses in Azure

- | | |
|---|--|
| <ul style="list-style-type: none">■ OS Standard■ OS Datacenter | <ul style="list-style-type: none">■ Each 2 CPU/16 Core license allows 2x8 Core or 1x16 Core■ Datacenter: May be used on-premise and in Azure simultaneously |
| <ul style="list-style-type: none">■ SQL Standard Core | <ul style="list-style-type: none">■ 1 Cloud Core for each 1 on-premise Core |
| <ul style="list-style-type: none">■ SQL Enterprise Core | <ul style="list-style-type: none">■ 4 Cloud Core for each 1 on-premises Core (Gen purpose)■ 1 Cloud Core for each 1 on-premises Core (Buss Critical) |

Design Optimization

- **Optimization is an ongoing effort !**
- **Keep innovating your landscape**
 - » Move Legacy applications to SaaS applications
 - » Move SQL servers to SQL as-a-service (IaaS to PaaS)
- **Cost best practices**
 - » Beware of VNET to VNET traffic; egress traffic is invoiced
 - » No egress cost using ExpressRoute local to on-premise
- **Credits for underuse of resource**
 - » B-series VM enables CPU bursting on accrued CPU credit
 - » Premium SSD P20 and smaller support bursting on accrued IOPS/Mbps credit

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Cost Management
Operations

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Auto shutdown

- Schedule resources to only run when needed
 - » For resources which are time bound (batch)
 - » For resources which do not require always on (dev/tst,acc)
- Shutdown resources at the end of the day, manual start
 - » For instance, Developer workstation environments
 - » Not applicable for Reserved Instances

Discussion

Can we define a rule of thumb for Auto shutdown?

Cleanup

- Shut down and decommission unused resources
- Clean up orphaned disks, network resources
- Clean up of backup and snapshots
- Clean up unused Public IP's
- Right-size redundancy

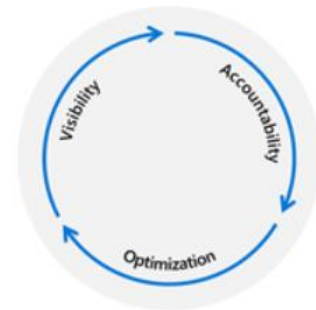


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Cost Management Governance Guidelines



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Cost Management Guidelines

- **Cost Manager**

- » Assign a global cost manager who handles overall budget and is responsible for the design for cost principle

- **Solution Design**

- » Make cost estimation a required part of the solution design phase

- **Budget tags**

- » Require budget tags on resource groups to compare between plan and actual

- **Benefits**

- » Cloud spend is in control
- » Accountable person who chases teams to design for cost
- » Visibility in budget versus actual spend

- **Continuous Improvement**

- » Measure your environment regularly for Cost optimization

Next Steps

- Customer “Homework” in this stage
 - » Who will be taking the Cost Manager role?
 - » Are current on-premise licenses re-usable in Azure?
 - » Think about adjusting or creating internal procedures for requesting resources in Azure?

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Discussion



A low-angle, upward-looking photograph of several tall skyscrapers. The image is heavily stylized with a red color overlay. The word "PROACT" is written in large, white, bold, sans-serif capital letters across the center of the image. The buildings are dark and their details are somewhat obscured by the red tint and the central text.

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